

# A Universal Field-Size Cap and Prize-Facing Completion Program for Mutual Correlated Agreement on Smooth Reed–Solomon Domains

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## Abstract

We prove a field-size-universal upper bound on the proximity parameter achievable in the grand mutual-correlated-agreement (MCA) challenge of Arnon–Boneh–Fenzi [ABF26]. Let  $\mathbb{F}$  be any finite field with  $|\mathbb{F}| < 2^{256}$ , let  $D$  be a smooth multiplicative evaluation domain of order  $n$  contained in a subfield  $B \subseteq \mathbb{F}$  (with  $B = \mathbb{F}$  allowed), and let  $C = \text{RS}[\mathbb{F}, D, k]$  have  $k = \rho n \leq 2^{40}$  and  $\rho \in \{1/2, 1/4, 1/8, 1/16\}$ . If  $2^{10} \mid n$  for  $\rho \in \{1/2, 1/4, 1/8\}$ , and  $2^{11} \mid n$  for  $\rho = 1/16$ , then

$$\varepsilon_{\text{mca}}(C, \delta) > \frac{1}{2k} \left(1 - \frac{n}{|\mathbb{F}|}\right) \geq 2^{-86} \gg 2^{-128}$$

for every  $\delta \in [1 - \rho - 2^{-9}, 1 - \rho)$  at rates  $1/2, 1/4, 1/8$ , and for every  $\delta \in [1 - \rho - 2^{-10}, 1 - \rho)$  at rate  $1/16$ . The same lower bound holds for  $\varepsilon_{\text{ca}}(C, \delta)$  on the integer-admissible subinterval ending at  $1 - \rho - 1/n$ . If  $|\mathbb{F}| \geq 2n$ , the displayed lower bound improves to  $2^{-42}$ . Consequently  $\delta_C^*(2^{-128}) \leq 1 - \rho - 2^{-9}$  for  $\rho \in \{1/2, 1/4, 1/8\}$  and  $\delta_C^*(2^{-128}) \leq 1 - \rho - 2^{-10}$  for  $\rho = 1/16$ , throughout the  $|\mathbb{F}| < 2^{256}$  challenge envelope. This is a prize-facing negative result: it does not determine the exact threshold, but it rules out an entire near-capacity band for every challenge-envelope row satisfying the printed divisor hypothesis.

The mechanism composes a pigeonhole lower bound on quotient-locator list fibers—a slack-two and subfield-pigeonhole sharpening of [Cho26a, Main Thm. (d)]—with a self-contained deep-point list-to-correlated-agreement conversion. This conversion is the same shallow consequence previously imported from Crites–Stewart [CS25, Thm. 2], but it is proved here directly with the integer-radius condition. The composition itself was first carried out, conditionally and with weaker constants, in the universal-cap theorem of [Cho26b] (error  $2^{-42}$  at gap  $2^{-11}$ , assuming  $|\mathbb{F}| \geq 2n$ ); the present note sharpens the gap, removes the field-size condition, and extends the cap to extension fields. No value-set or equidistribution input is used, so the per-fiber collision problem isolated in [Cho26b] is routed around rather than solved. We instantiate the bound at the KoalaBear-sextic parameters of [ABF26, §6.3], including the interleaved rows of their Tables 2–3, give an independent slacked variant through [BCHKS25, Thm. 1.9], and prove a two-radius refinement of the subfield-confinement theorem of [Cho26b], showing that over extension fields every explicit base-rational locator line is inert: the conversion-certified MCA failure must be witnessed by genuinely  $\mathbb{F}$ -valued lines.

# 1 Introduction

Proximity testing for Reed–Solomon codes underlies the soundness of modern SNARKs, and the sharpest available soundness accounting is phrased through *mutual correlated agreement* (MCA): a random linear combination  $f_1 + \gamma f_2$  being  $\delta$ -close to the code should force  $f_1, f_2$  to be simultaneously explained by codewords on a *common* agreement set, except with small probability  $\varepsilon_{\text{mca}}(C, \delta)$  over  $\gamma$ . The survey of Arnon, Boneh and Fenzi [ABF26] crystallizes the state of the art into a *grand MCA challenge*: for  $C = \text{RS}[\mathbb{F}, L, k]$  over a smooth domain  $L$ , with rate  $\rho \in \{1/2, 1/4, 1/8, 1/16\}$ , degree  $k \leq 2^{40}$ , and field size  $|\mathbb{F}| < 2^{256}$ , determine the largest proximity parameter  $\delta^*$  at which  $\varepsilon_{\text{mca}}(C, \delta) \leq 2^{-128}$  can hold.

Positive results reach the Johnson bound  $1 - \sqrt{\rho}$ . On the negative side, [Cho26b] showed  $\delta_C^*(2^{-128}) < 1 - \rho - 1/64$  for all prime fields of size up to roughly  $2^{150}$  (per-rate reaches between  $2^{150.6}$  and  $2^{161.8}$ ), by exhibiting quotient-locator witnesses whose bad-slope sets are value sets of restricted symmetric sums over small subgroups; it further proved a *conditional* universal cap over every field of size at most  $2^{256}$  — error  $2^{-42}$  at gap  $2^{-11}$ , assuming  $|\mathbb{F}| \geq 2n$  — by composing quotient-core lists with the same conversion isolated and proved directly here (its universal-cap theorem). Above the value-set reach, what remained open was the *error-one* question, identified there with the *per-fiber collision problem*: controlling, at a *fixed* large prime, the collisions of characteristic-zero locator values under reduction. Extension fields, the setting actually deployed ([ABF26, §6.3] works over a sextic extension of KoalaBear), posed an additional obstruction: by the subfield-confinement theorem of [Cho26b], refined in Section 6, every explicit witness of [Cho26a, Cho26b] is  $B$ -rational and contributes only  $p^{-5} < 2^{-154}$  over the deployed extension.

This note sharpens the universal cap — gap  $2^{-9}$  (resp.  $2^{-10}$ ) in place of  $2^{-11}$ , error  $2^{-86}$  with no condition relating  $n$  and  $|\mathbb{F}|$ , and extension fields handled directly — for every field below  $2^{256}$  at once, by changing what is counted.

**Theorem 1.1** (informal; see Theorem 4.1 and Corollary 5.1). *Let  $\mathbb{F}$  be any finite field with  $|\mathbb{F}| < 2^{256}$ , let  $D \subseteq \mathbb{F}^\times$  be a smooth multiplicative evaluation domain of order  $n$ , and let  $C = \text{RS}[\mathbb{F}, D, k]$  have  $k = \rho n \leq 2^{40}$  with  $\rho \in \{1/2, 1/4, 1/8, 1/16\}$ . Assume  $2^{10} \mid n$  for  $\rho \in \{1/2, 1/4, 1/8\}$ , and  $2^{11} \mid n$  for  $\rho = 1/16$ . Then*

$$\varepsilon_{\text{mca}}(C, \delta) > 2^{-86} \gg 2^{-128}$$

*for every  $\delta \in [1 - \rho - 2^{-9}, 1 - \rho]$  at rates  $1/2, 1/4, 1/8$ , and for every  $\delta \in [1 - \rho - 2^{-10}, 1 - \rho]$  at rate  $1/16$ . The same lower bound holds for  $\varepsilon_{\text{ca}}(C, \delta)$  on the integer-admissible subinterval ending at  $1 - \rho - 1/n$ . If  $|\mathbb{F}| \geq 2n$ , the lower bound is  $\geq 2^{-42}$ . Hence  $\delta_C^*(2^{-128}) \leq 1 - \rho - 2^{-9}$  for  $\rho \in \{1/2, 1/4, 1/8\}$  and  $\delta_C^*(2^{-128}) \leq 1 - \rho - 2^{-10}$  for  $\rho = 1/16$ .*

The proof composes two ingredients. The first (Lemma 3.1) is a pigeonhole lower bound on list sizes at a single received word: the quotient locators of [Cho26a] attach to each  $\ell$ -subset  $A$  of the order- $N$  quotient coset  $Q = D^a$  a codeword  $c_A$  at distance  $\leq 1 - \rho - t/N$  from the word  $u_{z_A} = x^{k+ta} + z_A x^{k+(t-1)a}$ , where  $z_A = -e_1(A)$ ; pigeonholing the  $\binom{N}{\ell}$  subsets over their  $\leq |B|$  slope values produces one word  $u_z$  carrying a list of size  $\binom{N}{\ell}/|B|$ . Two refinements of [Cho26a, Main Thm. (d)] matter here: we run the construction at *slack two* ( $t = 2$ ), so that the listed codewords have degree  $\leq k$  and live in  $\text{RS}[\mathbb{F}, D, k+1]$  without any divisibility demand on  $k+1$  (which is odd in all challenge instantiations, while  $a$  is a 2-power); and we pigeonhole over the *subfield*  $B$  containing  $D$  rather than over  $\mathbb{F}$ , which is what makes the extension-field case go through. The second ingredient is the direct deep-point conversion of Theorem 2.6: in the integer-error range  $f \leq n - k - 1$ , if  $\varepsilon_{\text{ca}}(C, \delta)$  is below  $\eta(\frac{1}{k} - \frac{n}{k|\mathbb{F}|})$ , then *every* list of  $\text{RS}[\mathbb{F}, D, k+1]$

at radius  $\delta$  has size at most  $\lceil |\mathbb{F}| \varepsilon_{\text{ca}} / (1 - \eta) \rceil$ . Taking  $\eta = \frac{1}{2}$ , the heavy fiber of Lemma 3.1 violates this whenever  $\binom{N}{\ell} \geq |B|(|\mathbb{F}|/k + 1)$  — an inequality with hundreds of bits of slack throughout the challenge envelope — and the contrapositive yields  $\varepsilon_{\text{ca}}(C, \delta) > \frac{1}{2k}(1 - n/|\mathbb{F}|)$  throughout the integer-admissible range. The MCA cap then persists up to capacity by support-wise MCA monotonicity.

No value-set counting, exponential-sum estimate, or equidistribution hypothesis enters: pigeonhole fibers exist over every field unconditionally, and Theorem 2.6 converts list mass into correlated-agreement error directly. The cost of this route, relative to the error-one results of [Cho26a, Cho26b] below  $2^{150}$ , is an error of order  $1/k$  rather than  $1 - o(1)$ , and a gap  $2^{-9}$  at the first three prize rates (uniformly  $2^{-10}$  across all four), rather than the fixed gap  $1/64$ ; Section 8 quantifies the trade-off and what remains open.

Beyond the universal cap (Corollary 5.1, with a two-tier summary combining it with [Cho26b]), we prove: a deployed-parameter corollary at the KoalaBear-sextic instantiation of [ABF26, §6.3], with  $\varepsilon_{\text{mca}} > 2^{-22}$  at gap  $2^{-7}$  and the same  $\varepsilon_{\text{ca}}$  lower bound on the integer-admissible subinterval (Corollary 5.4), extended to every interleaved row of [ABF26, Tables 2–3] via an interleaving-transfer lemma (Lemma 3.9 and Corollary 5.5); an independent slacked variant through [BCHKS25, Thm. 1.9] for robustness (Proposition 5.6); and a subfield-confinement lemma (Lemma 6.1) showing that for lines with  $B$ -valued words over an extension  $\mathbb{F}/B$ , every CA- or MCA-bad slope lies in  $B$ . The latter has a structural consequence (Corollary 6.2): at deployed parameters the lines certifying Corollary 5.4 are necessarily *not*  $B$ -rational, so the failure proved here is fiber-borne and currently non-constructive; exhibiting explicit certifying lines is posed as Open Problem 8.1.

**Roadmap.** Section 2 fixes definitions, aligned with [ABF26, Defs. 4.1, 4.3], proves the direct deep-point conversion used for the main cap, and records the separate BCHKS/ABF slacked fallback. Section 3 proves the fiber and interleaving lemmas, Section 4 the main theorem, Section 5 the challenge-level and deployed corollaries, and Section 6 subfield confinement. The new Section 7 states exactly what this paper contributes toward the Proximity Prize [PP26], gives conditional completion theorems for the MCA and interleaved-list thresholds, and isolates the proof obligations that would turn the present cap into a full threshold determination. Section 8 discusses scope, limits, calibration against the conjectural picture of [Cho26b], and open problems. Throughout, companion results of [Cho26a, Cho26b] are cited by their printed theorem names rather than numbers, since the numbering of those manuscripts is not yet frozen.

## 2 Preliminaries

**Codes and distances.** Throughout,  $B \subseteq \mathbb{F}$  are finite fields,  $q := |\mathbb{F}|$ , and  $D = \alpha H \subseteq B^\times$  is a multiplicative coset of a cyclic subgroup  $H$  of order  $n$ ; we write  $D$  for the evaluation domain (denoted  $L$  in [ABF26]). The subgroup case is  $\alpha = 1$ , and taking  $B = \mathbb{F}$  is always allowed. For  $k \leq n$ ,

$$\text{RS}[\mathbb{F}, D, k] := \{(p(x))_{x \in D} : p \in \mathbb{F}[X], \deg p < k\} \subseteq \mathbb{F}^n, \quad \rho := k/n.$$

For  $u, v \in \mathbb{F}^n$ ,  $\Delta(u, v)$  is the relative Hamming distance,  $\Delta(u, C) := \min_{c \in C} \Delta(u, c)$ , and the relative minimum distance of  $C = \text{RS}[\mathbb{F}, D, k]$  is  $\delta_{\min}(C) = 1 - \rho + 1/n$ . For  $S \subseteq D$ ,  $\Delta_S(u, v)$  and  $\Delta_S(u, C)$  denote the corresponding restricted relative distances on  $S$ . Lists are denoted

$$\Lambda(C, \delta, u) := \{c \in C : \Delta(u, c) \leq \delta\}, \quad \Lambda(C, \delta) := \max_{u \in \mathbb{F}^n} |\Lambda(C, \delta, u)|.$$

For  $s \geq 1$  the  $s$ -interleaved code is  $C^{\equiv s} := \{(c^{(1)}, \dots, c^{(s)}) : c^{(i)} \in C\} \subseteq (\mathbb{F}^n)^s$  with the column distance  $\Delta_s(F, G) := \frac{1}{n} |\{j : \exists i, f_j^{(i)} \neq g_j^{(i)}\}|$ , and  $\Delta_s(F, C^{\equiv s}) := \min_{\mathbf{c} \in C^{\equiv s}} \Delta_s(F, \mathbf{c})$ .

**Definition 2.1** (correlated agreement error; [ABF26, Def. 4.1] up to notation). For  $0 \leq \delta_{\text{fld}} \leq \delta_{\text{int}} < 1$ ,

$$\varepsilon_{\text{ca}}(C, \delta_{\text{fld}}, \delta_{\text{int}}) := \max_{f_1, f_2 \in \mathbb{F}^n} \Pr_{\gamma \leftarrow \mathbb{F}} \left[ \Delta(f_1 + \gamma f_2, C) \leq \delta_{\text{fld}} \wedge \Delta_2((f_1, f_2), C^{\equiv 2}) > \delta_{\text{int}} \right],$$

and the no-proximity-loss version is  $\varepsilon_{\text{ca}}(C, \delta) := \varepsilon_{\text{ca}}(C, \delta, \delta)$ . A slope  $\gamma$  realizing the event for a fixed pair  $(f_1, f_2)$  is called *CA-bad* for that pair.

**Definition 2.2** (mutual correlated agreement error, support-wise; [ABF26, Def. 4.3] up to notation). For  $\delta \in (0, 1)$ ,

$$\varepsilon_{\text{mca}}(C, \delta) := \max_{f_1, f_2 \in \mathbb{F}^n} \Pr_{\gamma \leftarrow \mathbb{F}} \left[ \begin{array}{l} \exists S \subseteq D, |S| \geq (1 - \delta)n, \Delta_S(f_1 + \gamma f_2, C) = 0, \\ \text{and } \Delta_S((f_1, f_2), C^{\equiv 2}) > 0 \end{array} \right].$$

Equivalently,  $\gamma$  is MCA-bad for  $(f_1, f_2)$  if some large agreement set  $S$  explains the point  $f_1 + \gamma f_2$ , but the same set  $S$  does not simultaneously explain  $f_1$  and  $f_2$  by two codewords of  $C$ .

**Definition 2.3** (challenge threshold). Following the grand MCA challenge of [ABF26, §1], for  $\varepsilon^* \in (0, 1)$  set  $\delta_C^*(\varepsilon^*) := \sup\{\delta \in (0, 1 - \rho) : \varepsilon_{\text{mca}}(C, \delta) \leq \varepsilon^*\}$ , with  $\sup \emptyset := 0$ .

**Fact 2.4** (cf. [ABF26, Fact 4.5]). *For every  $\delta \in (0, 1)$ :  $\varepsilon_{\text{ca}}(C, \delta) \leq \varepsilon_{\text{mca}}(C, \delta)$ .*

*Proof.* Fix  $(f_1, f_2)$  and let  $\gamma$  be CA-bad:  $\Delta(f_1 + \gamma f_2, C) \leq \delta$  and  $\Delta_2((f_1, f_2), C^{\equiv 2}) > \delta$ . Choose  $S \subseteq D$  of size at least  $(1 - \delta)n$  on which  $f_1 + \gamma f_2$  agrees with a codeword of  $C$ . Since  $(f_1, f_2)$  is not  $\delta$ -close to  $C^{\equiv 2}$ , there is no set of size at least  $(1 - \delta)n$  on which  $f_1$  and  $f_2$  are simultaneously explained by codewords of  $C$ ; in particular this fails on the chosen  $S$ . Thus  $\gamma$  is MCA-bad for the same pair. Taking maxima over pairs gives the claim.  $\square$

**Lemma 2.5** (support-wise MCA monotonicity). *For every linear code  $C \subseteq \mathbb{F}^D$ , the function*

$$\delta \mapsto \varepsilon_{\text{mca}}(C, \delta)$$

*is nondecreasing.*

*Proof.* Fix a pair  $(f_1, f_2)$  and suppose that  $\gamma$  is MCA-bad at radius  $\delta$ . Then there is a set  $S \subseteq D$  with

$$|S| \geq (1 - \delta)n$$

such that  $f_1 + \gamma f_2$  is explained by  $C$  on  $S$ , but  $(f_1, f_2)$  is not simultaneously explained by two codewords of  $C$  on the same  $S$ . If  $\delta' \geq \delta$ , then

$$|S| \geq (1 - \delta)n \geq (1 - \delta')n,$$

so the same set  $S$  witnesses that  $\gamma$  is MCA-bad at radius  $\delta'$ . Taking the maximum over pairs gives the claim.  $\square$

**Conversion theorem.** The main composition below uses the deep-point list-to-CA conversion proved here directly. This is the same consequence previously imported from Crites–Stewart, with the integer-radius admissibility condition made explicit; see Remark 2.9. We keep the slacked BCHKS/ABF fallback as a separate imported route.

**Theorem 2.6** (deep-point list-to-CA conversion). *Let  $C = \text{RS}[\mathbb{F}, D, k]$ , let  $C^+ := \text{RS}[\mathbb{F}, D, k + 1]$ , let  $q := |\mathbb{F}|$  and  $n := |D|$ , and assume  $q > n$ . Let  $\delta \in (0, 1)$  and set  $f_\delta := \lfloor \delta n \rfloor$ . Assume*

$$f_\delta \leq n - k - 1.$$

*For any  $\eta \in [0, 1)$ , if*

$$\varepsilon_{\text{ca}}(C, \delta) \leq \eta \left( \frac{1}{k} - \frac{n}{kq} \right),$$

*then*

$$\Lambda(C^+, \delta) \leq \left\lceil \frac{q \varepsilon_{\text{ca}}(C, \delta)}{1 - \eta} \right\rceil.$$

*Proof.* Put  $\epsilon := \varepsilon_{\text{ca}}(C, \delta)$  and  $f = f_\delta$ . Let  $U : D \rightarrow \mathbb{F}$  be any received word and suppose that  $P_1, \dots, P_L \in \mathbb{F}[X]_{<k+1}$  are distinct polynomials whose restrictions to  $D$  all lie in the Hamming ball of relative radius  $\delta$  around  $U$ . Since distances are integral, each  $P_i$  agrees with  $U$  on at least

$$a := n - f$$

points of  $D$ . The hypothesis  $f \leq n - k - 1$  gives  $a \geq k + 1$ , in particular  $a > k$ .

Let  $\Omega := \mathbb{F} \setminus D$ , so  $|\Omega| = q - n$ . For each  $\alpha \in \Omega$  define the simple-pole line

$$f_\alpha(x) = \frac{U(x)}{x - \alpha}, \quad g_\alpha(x) = -\frac{1}{x - \alpha} \quad (x \in D).$$

If  $P_i$  agrees with  $U$  on a set  $S_i$  of size at least  $a$ , then for  $z_i(\alpha) := P_i(\alpha)$  and every  $x \in S_i$ ,

$$f_\alpha(x) + z_i(\alpha)g_\alpha(x) = \frac{P_i(x) - P_i(\alpha)}{x - \alpha}.$$

The right-hand side is the restriction of a polynomial of degree  $< k$ . Thus each distinct value among the  $P_i(\alpha)$  gives a slope for which the line point is  $\delta$ -close to  $C$ .

We next check the correlated-agreement far condition. If a polynomial  $G \in \mathbb{F}[X]_{<k}$  agreed with  $g_\alpha$  on more than  $k$  points of  $D$ , then

$$(X - \alpha)G(X) + 1$$

would be a polynomial of degree at most  $k$  with more than  $k$  roots, but its value at  $X = \alpha$  is 1, a contradiction. Hence  $(f_\alpha, g_\alpha)$  is not  $\delta$ -close to  $C^{\equiv 2}$ , because any common explanation on a support of size at least  $a > k$  would in particular explain  $g_\alpha$  there. Therefore the distinct values of  $P_i(\alpha)$  are CA-bad slopes.

It remains to lower-bound the number of distinct values for some  $\alpha$ . For  $i \neq j$ , the polynomial  $P_i - P_j$  is nonzero of degree at most  $k$ , so it has at most  $k$  roots in  $\Omega$ . Let  $M(\alpha)$  be the number of distinct values among  $P_1(\alpha), \dots, P_L(\alpha)$ . Over all  $\alpha \in \Omega$ , the total number of colliding unordered pairs  $(i, j)$  with  $P_i(\alpha) = P_j(\alpha)$  is at most  $k \binom{L}{2}$ . Equivalently, if the fiber sizes of the map  $i \mapsto P_i(\alpha)$  are  $m_v(\alpha)$ , then

$$\sum_{\alpha \in \Omega} \sum_v \binom{m_v(\alpha)}{2} \leq k \binom{L}{2}.$$

Choose  $\alpha \in \Omega$  for which the inner collision count is at most  $k\binom{L}{2}/(q-n)$ . If the fiber sizes of the map  $i \mapsto P_i(\alpha)$  are  $m_1, \dots, m_{M(\alpha)}$ , then

$$\sum_r m_r = L, \quad \sum_r m_r^2 = L + 2 \sum_r \binom{m_r}{2} \leq L + \frac{kL(L-1)}{q-n}.$$

Cauchy–Schwarz therefore gives

$$L^2 \leq M(\alpha) \sum_r m_r^2, \quad M(\alpha) \geq \frac{L(q-n)}{q-n+k(L-1)} \geq \frac{L(q-n)}{q-n+kL}.$$

Consequently

$$\epsilon \geq \frac{1}{q} \cdot \frac{L(q-n)}{q-n+kL}.$$

Solving this inequality for  $L$  gives

$$L \leq \frac{\epsilon q(q-n)}{q-n-k\epsilon q},$$

provided  $\epsilon < (q-n)/(kq)$ . Under the stated hypothesis,  $\epsilon \leq \eta(q-n)/(kq)$  with  $\eta < 1$ , so the denominator is at least  $(1-\eta)(q-n)$  and hence

$$L \leq \frac{q\epsilon}{1-\eta}.$$

Since  $U$  and the list were arbitrary, the displayed ceiling bounds the maximum list size of  $C^+$  at radius  $\delta$ .  $\square$

*Remark 2.7* (relation to Crites–Stewart). The displayed conversion is the only consequence of Crites–Stewart needed by the universal-cap argument. Earlier drafts imported it from [CS25, Thm. 2]; the proof above is included to make the main cap self-contained. The separate slacked fallback through BCHKS/ABF remains an independent imported route.

**Theorem 2.8** ([BCHKS25, Thm. 1.9], as stated in [ABF26, Thm. 5.2]). *Let  $C = \text{RS}[\mathbb{F}, D, k]$ , let  $\rho = k/n$ , and let  $\delta \in (0, 1-\rho)$  satisfy  $\delta + 2/n \leq 1-\rho - 1/n$ . If*

$$\varepsilon_{\text{ca}}\left(C, \delta + \frac{2}{n}, 1-\rho - \frac{1}{n}\right) < \frac{1}{2n}, \quad \text{then} \quad \Lambda(C, \delta) < |\mathbb{F}|.$$

*Remark 2.9* (on imports and radius conventions). Theorem 2.6 is the only Crites–Stewart-type consequence used by the main cap, and it is now proved above by the simple-pole deep-point argument. The important admissibility condition is the integer-radius condition

$$\lfloor \delta n \rfloor \leq n - k - 1.$$

Consequently, the no-loss CA lower bound proved below should be read only in this admissible range. The Proximity-Prize MCA cap is unaffected: we prove the lower bound at the endpoint

$$\delta_N := 1 - \rho - \frac{2}{N}$$

and then use support-wise MCA monotonicity to extend the MCA failure to all larger sub-capacity radii.

Theorem 2.8 remains a separate slacked fallback imported through the BCHKS/ABF route. It is not needed for the main field-size-universal MCA cap.

### 3 Locator fibers and interleaving transfer

Fix  $N \mid n$ , set  $a := n/N$ , and let  $Q := D^a := \{x^a : x \in D\} \subseteq B^\times$ , a multiplicative coset of order  $N$ . The  $a$ -th power map  $D \rightarrow Q$  is surjective and its fibers  $S_b := \{x \in D : x^a = b\}$  have exactly  $a$  elements each. Since  $a \mid n \mid |B| - 1$ , the characteristic of  $\mathbb{F}$  does not divide  $a$ , so  $X^a - b$  is separable for  $b \neq 0$  and its full root set inside  $D$  is  $S_b$  whenever  $b \in Q$ . For a set  $A$ ,  $e_j(A)$  denotes the  $j$ -th elementary symmetric function of its elements.

**Lemma 3.1** (locator fibers are lists). *Let  $B \subseteq \mathbb{F}$ , let  $D \subseteq B^\times$  be a multiplicative coset of order  $n$ , and let  $k$  satisfy  $a \mid k$  and  $\rho = k/n$ . Write  $\ell_1 := \rho N + 1$  and  $\ell_2 := \rho N + 2$ .*

(i) *If  $\ell_1 \leq N$ , there exists  $z \in B$  such that, with  $u_z := (x^{k+a} + z x^k)_{x \in D} \in \mathbb{F}^n$ ,*

$$|\Lambda(\text{RS}[\mathbb{F}, D, k], 1 - \rho - \frac{1}{N}, u_z)| \geq \binom{N}{\ell_1} / |B|.$$

(ii) *If  $\ell_2 \leq N$ , there exists  $z \in B$  such that, with  $u_z := (x^{k+2a} + z x^{k+a})_{x \in D} \in \mathbb{F}^n$ ,*

$$|\Lambda(\text{RS}[\mathbb{F}, D, k+1], 1 - \rho - \frac{2}{N}, u_z)| \geq \binom{N}{\ell_2} / |B|.$$

*In both cases the words  $u_z$  are  $B$ -valued and all listed codewords lie in  $\text{RS}[B, D, k]$  (resp.  $\text{RS}[B, D, k+1]$ ).*

*Proof.* We prove (ii); part (i) is the identical computation one degree rung lower and is [Cho26a, Main Thm. (d)] sharpened by pigeonholing over  $B$  in place of  $\mathbb{F}$ .

Let  $A \subseteq Q$  with  $|A| = \ell_2$ , and put

$$L_A(X) := \prod_{b \in A} (X^a - b) = \sum_{j=0}^{\ell_2} (-1)^j e_j(A) X^{a(\ell_2-j)} = X^{k+2a} - e_1(A) X^{k+a} + R_A(X),$$

where  $a\ell_2 = k + 2a$ ,  $a(\ell_2 - 1) = k + a$ , and  $R_A$  collects the terms with  $j \geq 2$ , so  $\deg R_A \leq a(\ell_2 - 2) = k$ . The polynomial  $L_A$  is squarefree with root set  $S_A := \bigcup_{b \in A} S_b \subseteq D$  of size  $a\ell_2 = k + 2a$ .

Set  $z_A := -e_1(A) \in B$  and  $c_A := (-R_A(x))_{x \in D}$ . Since  $\deg R_A \leq k$ , we have  $c_A \in \text{RS}[B, D, k+1] \subseteq \text{RS}[\mathbb{F}, D, k+1]$ . For every  $x \in S_A$ ,

$$0 = L_A(x) = x^{k+2a} + z_A x^{k+a} + R_A(x), \quad \text{i.e.} \quad u_{z_A}(x) = c_A(x),$$

so  $u_{z_A}$  and  $c_A$  agree on at least  $k + 2a$  points of  $D$  and  $\Delta(u_{z_A}, c_A) \leq 1 - \rho - 2/N$ .

The map  $A \mapsto c_A$  is injective on the  $\ell_2$ -subsets of  $Q$  sharing a common slope value: if  $z_A = z_{A'}$  and  $c_A = c_{A'}$ , then  $R_A$  and  $R_{A'}$  are polynomials of degree  $\leq k < n$  agreeing on all  $n$  points of  $D$ , hence equal as polynomials; then  $L_A = L_{A'}$ , so  $S_A = S_{A'}$ , and  $A = \{x^a : x \in S_A\} = A'$ .

Finally,  $z_A = -e_1(A)$  takes at most  $|B|$  values as  $A$  ranges over the  $\binom{N}{\ell_2}$  subsets; some  $z \in B$  is attained by at least  $\binom{N}{\ell_2} / |B|$  of them, and the corresponding codewords  $c_A$  are pairwise distinct elements of  $\Lambda(\text{RS}[\mathbb{F}, D, k+1], 1 - \rho - 2/N, u_z)$ .  $\square$



*Remark 3.2* (why slack two). Part (i) applied to  $C^+ = \text{RS}[\mathbb{F}, D, k+1]$  directly would require  $a \mid k+1$ ; in every challenge instantiation  $a$  is a power of two and  $k+1$  is odd, so this fails. At slack two the listed codewords have degree  $\leq k$ , i.e. lie in  $C^+$ , while the divisibility demand stays on  $k$ . This is exactly the interface needed by Theorem 2.6, whose list conclusion concerns  $C^+$  while its hypothesis concerns  $C$ .

*Remark 3.3* (the subfield pigeonhole elsewhere). The same one-line strengthening — pigeonholing locator data over the field of definition  $B$  rather than over  $\mathbb{F}$  — also upgrades the coefficient-pigeonhole bound of [Cho26b] to the generated field: the prefix map  $\Phi_\sigma$  there has image in  $B^\sigma$  whenever  $D \subseteq B$ , since the elementary symmetric prefixes of subsets of  $D$  are  $B$ -valued. This is the form consumed by the “necessary” half of the list-profile conjecture of [Cho26b], which states the entropy floor at the generated field  $\tau^*(\rho, q_D)$ ; Lemma 3.1 supplies the missing subfield step.

*Remark 3.4* (lists, not lines). The words  $u_z$  of Lemma 3.1 lie on the  $B$ -rational line  $f + zg$  with  $f = (x^{k+ta})_{x \in D}$ ,  $g = (x^{k+(t-1)a})_{x \in D}$ , but the lemma is purely a statement about lists at a received word and no line structure is used in Section 4. This matters: by Corollary 6.2 below, over a proper extension  $\mathbb{F} \supsetneq B$  the lines that end up certifying the CA failure cannot be these (or any)  $B$ -rational lines.

### 3.1 Quotient-profile floors with remainder supports

The fiber lemma above uses supports that are unions of complete quotient fibers. For threshold work one also needs agreements that are not multiples of the fiber size. The following locator-prefix pigeonhole gives a finite lower ledger for such remainder supports. It is a lower-floor theorem only; the matching upper ledger over all divisor scales remains part of the prize-completion program.

Let  $c \mid n$  and put  $Q_c := D^c$ , so the map

$$\pi_c : D \rightarrow Q_c, \quad x \mapsto x^c$$

has fibers of size  $c$  and  $|Q_c| = N = n/c$ . For  $0 \leq s < c$  and an integer  $\sigma \geq 0$ , define the prefix weight

$$w_c(s, \sigma) := \#\{h : 1 \leq h \leq \sigma, h \bmod c \in \{0, 1, \dots, s\}\},$$

where residues are taken in  $\{0, 1, \dots, c-1\}$ . Thus  $w_c(0, \sigma) = \lfloor \sigma/c \rfloor$  for full-fiber supports.

**Lemma 3.5** (quotient-remainder locator-prefix lower floor). *Let  $B \subseteq \mathbb{F}$ , let  $D \subseteq B^\times$  be a multiplicative coset of order  $n$ , and let  $K < n$ . Fix  $c \mid n$ , write  $N = n/c$ , and let*

$$A_0 = mc + s, \quad 0 \leq s < c, \quad 0 \leq m \leq N, \quad A_0 \geq K.$$

*If  $s > 0$ , assume  $mc + s \leq n$ . Put  $\sigma = A_0 - K$  and*

$$M_{c,m,s} := \binom{N}{m} \binom{n-mc}{s}.$$

*Then there is a  $B$ -valued received word  $U : D \rightarrow B$  such that*

$$|\Lambda(\text{RS}[\mathbb{F}, D, K], 1 - A_0/n, U)| \geq \left\lceil \frac{M_{c,m,s}}{|B|^{w_c(s, \sigma)}} \right\rceil.$$

*The listed codewords may be chosen in  $\text{RS}[B, D, K]$ .*



*Proof.* For  $M \subseteq Q_c$  with  $|M| = m$ , write

$$P_M(X) = \prod_{b \in M} (X^c - b).$$

For  $R \subseteq D \setminus \pi_c^{-1}(M)$  with  $|R| = s$ , put

$$E_R(X) = \prod_{r \in R} (X - r), \quad L_{M,R}(X) = P_M(X)E_R(X).$$

Then  $L_{M,R}$  is a squarefree locator of degree  $A_0$  whose roots are exactly  $\pi_c^{-1}(M) \cup R$ . Write

$$L_{M,R}(X) = X^{A_0} + \sum_{h=1}^{A_0} \lambda_h(M, R) X^{A_0-h}.$$

The coefficients lie in  $B$ . Moreover, because  $P_M$  has monomials only in degrees congruent to 0 modulo  $c$  and  $E_R$  has degree  $s$ , the coefficient  $\lambda_h(M, R)$  is zero unless  $h \bmod c \in \{0, 1, \dots, s\}$ . Consequently the high-degree prefix

$$(\lambda_h(M, R) : 1 \leq h \leq \sigma, h \bmod c \in \{0, 1, \dots, s\})$$

takes at most  $|B|^{w_c(s, \sigma)}$  values.

Pigeonhole over these prefix values. There is a fixed prefix vector  $\xi$  and a subfamily  $\mathcal{T}_\xi$  of at least  $M_{c,m,s}/|B|^{w_c(s, \sigma)}$  pairs  $(M, R)$  with this prefix. Let

$$U_\xi(X) = X^{A_0} + \sum_{1 \leq h \leq \sigma} \lambda_h(\xi) X^{A_0-h}$$

be the corresponding high-degree part, restricted to  $D$ . For every  $(M, R) \in \mathcal{T}_\xi$ , the difference

$$R_{M,R}(X) := L_{M,R}(X) - U_\xi(X)$$

has degree  $< K$ . Hence the codeword  $-R_{M,R}$  agrees with  $U_\xi$  on all roots of  $L_{M,R}$ , a set of size  $A_0$ .

It remains only to check that the listed codewords are distinct. If two pairs  $(M, R)$  and  $(M', R')$  in the same prefix class gave the same codeword on  $D$ , then the two remainders of degree  $< K < n$  would be equal as polynomials. Since the high prefixes are also equal,  $L_{M,R} = L_{M',R'}$ . Their root sets in  $D$  are therefore equal. A full fiber has size  $c$  while  $|R| = s < c$ , so the full fibers contained in this root set are exactly  $M$ , and then the residual set is forced as well. Thus  $(M, R) = (M', R')$ .  $\square$

**Corollary 3.6** (augmented slack-one quotient floor). *Let  $B \subseteq \mathbb{F}$  and let  $D \subseteq B^\times$  be a multiplicative coset of order  $n$ . Put  $C = \text{RS}[\mathbb{F}, D, k]$  and  $C^+ = \text{RS}[\mathbb{F}, D, k+1]$ . If  $c \mid \gcd(n, k)$ , then there is a  $B$ -valued received word  $U_c$  such that*

$$|\Lambda(C^+, 1 - (k+c)/n, U_c)| \geq \binom{n/c}{k/c+1}.$$

*One may take  $U_c(x) = x^{k+c}$ . In particular, for  $c = 1$  this gives the universal first-grid floor*

$$|\Lambda(C^+, 1 - (k+1)/n, x^{k+1})| \geq \binom{n}{k+1},$$

*with no quotient denominator. The  $c = 1$  statement in fact holds for any set  $D \subseteq B$  of  $n$  distinct points, not only for multiplicative cosets.*

*Proof.* For  $c > 1$ , apply Lemma 3.5 with  $K = k + 1$ ,  $s = 0$ , and  $m = k/c + 1$ . Then  $A_0 = k + c$  and  $\sigma = A_0 - K = c - 1$ , so  $w_c(0, c - 1) = 0$ . The high-degree prefix is only the monomial  $X^{k+c}$ , giving the displayed word. For  $c = 1$ , the same argument is just the ordinary locator identity for all  $(k + 1)$ -subsets of  $D$ : the high-degree prefix of a monic locator of degree  $k + 1$  is  $X^{k+1}$ , and its remainder has degree at most  $k$ .  $\square$

**Corollary 3.7** (first-grid deep-point cap). *Let  $C = \text{RS}[\mathbb{F}, D, k]$ ,  $q = |\mathbb{F}|$ , and suppose  $q > n$ . Let  $c \mid \gcd(n, k)$ . If  $c > 1$ , assume  $D \subseteq B^\times$  is a multiplicative coset as above; for  $c = 1$ , no smoothness is needed. If*

$$\binom{n/c}{k/c + 1} \geq q/k + 1,$$

*then*

$$\varepsilon_{\text{ca}}(C, 1 - (k + c)/n) > \frac{1}{2k} \left(1 - \frac{n}{q}\right), \quad \varepsilon_{\text{mca}}(C, 1 - (k + c)/n) > \frac{1}{2k} \left(1 - \frac{n}{q}\right).$$

*For  $c = 1$  this is the first closed grid point below capacity.*

*Proof.* Corollary 3.6 gives a  $C^+$ -list of size  $L \geq q/k + 1$  at agreement  $k + c$ . The proof of the deep-point conversion Theorem 2.6 applies whenever the agreement is  $> k$ , so it applies here for every  $c \geq 1$ . If the displayed CA lower bound failed, then

$$\varepsilon_{\text{ca}}(C, 1 - (k + c)/n) \leq \frac{q - n}{2kq}.$$

Theorem 2.6 with  $\eta = 1/2$  would give

$$\Lambda(C^+, 1 - (k + c)/n) \leq \left\lceil \frac{q - n}{k} \right\rceil < q/k + 1,$$

contradicting the list lower bound. Every no-loss CA-bad slope is also support-wise MCA-bad on the same explaining support, giving the MCA inequality.  $\square$

**Corollary 3.8** (first-grid cap for large challenge-envelope rows). *Let  $\rho \in \{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}\}$ , let  $C = \text{RS}[\mathbb{F}, D, k]$  with  $|D| = n$ ,  $k = \rho n$ ,  $q = |\mathbb{F}| < 2^{256}$ , and  $q > n$ . Assume*

$$\frac{\rho}{k} \geq \left| \begin{array}{c|ccc} \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & \frac{1}{16} \\ \hline 127 & 78 & 58 & 47 \end{array} \right|$$

*for the corresponding rate. Then the first closed grid point below capacity is already CA- and MCA-unsafe:*

$$\varepsilon_{\text{ca}}(C, 1 - \rho - 1/n) > \frac{1}{2k} \left(1 - \frac{n}{q}\right), \quad \varepsilon_{\text{mca}}(C, 1 - \rho - 1/n) > \frac{1}{2k} \left(1 - \frac{n}{q}\right).$$

*If also  $k \leq 2^{40}$ , then this lower bound is  $> 2^{-86}$  and hence is far above the prize target  $2^{-128}$ .*

*Proof.* By Corollary 3.7 with  $c = 1$ , it is enough to check

$$\binom{n}{k + 1} \geq q/k + 1.$$

Since  $q < 2^{256}$ , the stronger sufficient condition is  $k(\binom{n}{k+1} - 1) \geq 2^{256}$ . The four printed lower bounds on  $k$  are exactly small finite checks of this inequality at the corresponding rate; the left side is increasing thereafter. For the error size,  $q \geq n + 1$ ,  $n = k/\rho \leq 16k$ , and  $k \leq 2^{40}$  give

$$\frac{1}{2k} \left(1 - \frac{n}{q}\right) \geq \frac{1}{2k(n+1)} > 2^{-86}.$$

□

**Lemma 3.9** (interleaving transfer). *For every linear code  $C \subseteq \mathbb{F}^n$ , every  $s \geq 1$  and every  $\delta \in (0, 1)$ :*

$$\varepsilon_{\text{ca}}(C^{\equiv s}, \delta) \geq \varepsilon_{\text{ca}}(C, \delta), \quad \varepsilon_{\text{mca}}(C^{\equiv s}, \delta) \geq \varepsilon_{\text{mca}}(C, \delta).$$

*Proof.* For  $h \in \mathbb{F}^n$  write  $h^\Delta := (h, \dots, h) \in (\mathbb{F}^n)^s$ . For any codeword tuple  $\mathbf{c} = (c^{(1)}, \dots, c^{(s)}) \in C^{\equiv s}$ , the columns where  $h^\Delta$  and  $\mathbf{c}$  agree are  $\bigcap_i \{j : c_j^{(i)} = h_j\} \subseteq \{j : c_j^{(1)} = h_j\}$ , so  $\Delta_s(h^\Delta, C^{\equiv s}) \geq \Delta(h, C)$ ; taking  $c^{(1)} = \dots = c^{(s)}$  gives equality:  $\Delta_s(h^\Delta, C^{\equiv s}) = \Delta(h, C)$ .

Fix  $(f_1, f_2)$  attaining  $\varepsilon_{\text{ca}}(C, \delta)$  (resp.  $\varepsilon_{\text{mca}}$ ) and consider the pair  $(f_1^\Delta, f_2^\Delta)$  for  $C^{\equiv s}$ , with the slope  $\gamma$  acting row-wise, so  $f_1^\Delta + \gamma f_2^\Delta = (f_1 + \gamma f_2)^\Delta$ . By the displayed equality,  $\Delta_s((f_1 + \gamma f_2)^\Delta, C^{\equiv s}) \leq \delta$  iff  $\Delta(f_1 + \gamma f_2, C) \leq \delta$ . For the correlated condition, the pair  $(f_1^\Delta, f_2^\Delta)$  viewed as a  $2s$ -row interleaved word has rows  $f_1$  ( $s$  times) and  $f_2$  ( $s$  times); by the same column argument its distance to  $(C^{\equiv s})^{\equiv 2} = C^{\equiv 2s}$  equals  $\Delta_2((f_1, f_2), C^{\equiv 2})$ . For the MCA condition, a common set  $S$  explains  $f_1^\Delta$  and  $f_2^\Delta$  iff it explains  $f_1$  and  $f_2$ . Hence  $\gamma$  is CA-bad (resp. MCA-bad) for  $(f_1^\Delta, f_2^\Delta)$  iff it is so for  $(f_1, f_2)$ , and the maxima over pairs can only grow. □

## 4 The main theorem

**Theorem 4.1** (universal cap). *Let  $B \subseteq \mathbb{F}$  be finite fields, let  $q = |B|$ , let  $D \subseteq B^\times$  be a multiplicative coset of order  $n$ , and let  $k$  have  $\rho = k/n \in (0, 1)$ ; set  $C := \text{RS}[\mathbb{F}, D, k]$ . Let  $N \mid n$  satisfy  $a := n/N \mid k$  and  $(1 - \rho)N \geq 3$ . If*

$$\binom{N}{\rho N + 2} \geq |B| \left(\frac{q}{k} + 1\right), \tag{1}$$

*then, writing*

$$\delta_N := 1 - \rho - \frac{2}{N},$$

*we have*

$$\varepsilon_{\text{ca}}(C, \delta) > \frac{1}{2k} \left(1 - \frac{n}{q}\right)$$

*for every integer-admissible radius*

$$\delta_N \leq \delta \leq 1 - \rho - \frac{1}{n}.$$

*In particular,*

$$\varepsilon_{\text{mca}}(C, \delta) > \frac{1}{2k} \left(1 - \frac{n}{q}\right)$$

*for every*

$$\delta_N \leq \delta < 1 - \rho.$$

Consequently,

$$\delta_C^*(\varepsilon^*) \leq 1 - \rho - \frac{2}{N}$$

for every

$$\varepsilon^* \leq \frac{1}{2k} \left(1 - \frac{n}{q}\right).$$

*Proof.* Note first that  $\rho N = k/a \in \mathbb{Z}$ ,  $\ell_2 = \rho N + 2 \leq N - 1$  by  $(1 - \rho)N \geq 3$ , and  $\delta_N = 1 - \rho - 2/N > 0$  for the same reason. Also  $k + 1 \leq n$ .

Fix

$$\delta \in [\delta_N, 1 - \rho - \frac{1}{n}].$$

Then

$$\lfloor \delta n \rfloor \leq n - k - 1,$$

so Theorem 2.6 is applicable.

Suppose toward a contradiction that

$$\varepsilon_{\text{ca}}(C, \delta) \leq \frac{1}{2k} \left(1 - \frac{n}{q}\right).$$

Apply Theorem 2.6 with  $\eta = \frac{1}{2}$ . Its hypothesis holds, so

$$\Lambda(C^+, \delta) \leq \left\lceil 2q \cdot \varepsilon_{\text{ca}}(C, \delta) \right\rceil \leq \left\lceil \frac{q - n}{k} \right\rceil < \frac{q - n}{k} + 1 \leq \frac{q}{k} + 1.$$

On the other hand, by Lemma 3.1(ii) and monotonicity of lists in the radius, since  $\delta \geq \delta_N$ ,

$$\Lambda(C^+, \delta) \geq |\Lambda(C^+, \delta_N, u_z)| \geq \binom{N}{\ell_2} / |B| \geq \frac{q}{k} + 1$$

by (1). This is a contradiction. Hence

$$\varepsilon_{\text{ca}}(C, \delta) > \frac{1}{2k} \left(1 - \frac{n}{q}\right)$$

for every  $\delta \in [\delta_N, 1 - \rho - 1/n]$ .

In particular, the above holds at  $\delta = \delta_N$ . By Fact 2.4,

$$\varepsilon_{\text{mca}}(C, \delta_N) \geq \varepsilon_{\text{ca}}(C, \delta_N) > \frac{1}{2k} \left(1 - \frac{n}{q}\right).$$

By support-wise MCA monotonicity, Lemma 2.5, the same lower bound for  $\varepsilon_{\text{mca}}(C, \delta)$  holds for every

$$\delta \in [\delta_N, 1 - \rho).$$

Thus no sub-capacity radius

$$\delta \geq 1 - \rho - \frac{2}{N}$$

is  $\varepsilon^*$ -admissible whenever

$$\varepsilon^* \leq \frac{1}{2k} \left(1 - \frac{n}{q}\right),$$

which proves the claimed bound on  $\delta_C^*(\varepsilon^*)$ . □

*Remark 4.2* (constants). Choosing  $\eta = 1 - \theta$  in Theorem 2.6 improves the conclusion to  $\varepsilon_{\text{ca}} > \frac{1-\theta}{k}(1 - \frac{n}{q})$  at the price of strengthening (1) to  $\binom{N}{\rho N+2} \geq |B|(\frac{q}{\theta k} + 1)$ ; given the hundreds of bits of slack in (1) throughout the challenge envelope (Table 1), the error constant is effectively  $(1 - o(1))/k$ . We fix  $\eta = \frac{1}{2}$  for cleanliness.

*Remark 4.3* (scope). No smoothness is assumed beyond the existence of a single divisor  $N \mid n$  with  $a = n/N \mid k$  and (1). The theorem applies verbatim to 2-adic, mixed-radix, or any other FFT-friendly multiplicative domain, and — via the subfield pigeonhole — to any extension field  $\mathbb{F} \supseteq B$  with  $D \subseteq B$ .

The deep-point conversion is an integer-radius theorem with the condition  $\lfloor \delta n \rfloor \leq n - k - 1$ . Therefore the no-loss CA lower bound is stated for

$$\delta \leq 1 - \rho - \frac{1}{n}.$$

The MCA threshold cap is stronger in the range needed for the Proximity Prize: once the support-wise MCA lower bound is proved at

$$\delta_N = 1 - \rho - \frac{2}{N},$$

monotonicity of support-wise MCA extends it to every larger  $\delta < 1 - \rho$ .

## 5 Consequences for the grand MCA challenge

**Corollary 5.1** (universal field-size cap for the challenge envelope). *Let  $\rho \in \{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}\}$  and set*

$$N_\rho := 1024 \quad (\rho \in \{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}\}), \quad N_{1/16} := 2048.$$

*Let  $\mathbb{F}$  be any finite field with  $q < 2^{256}$ , let  $B \subseteq \mathbb{F}$  be any subfield, let  $D \subseteq B^\times$  be a multiplicative coset of order  $n$  with  $N_\rho \mid n$ , and let  $k = \rho n \leq 2^{40}$ . Then  $C = \text{RS}[\mathbb{F}, D, k]$  satisfies*

$$\varepsilon_{\text{mca}}(C, \delta) > \frac{1}{2k} \left(1 - \frac{n}{q}\right) \geq 2^{-86} \gg 2^{-128}$$

*for every*

$$\delta \in \left[1 - \rho - \frac{2}{N_\rho}, 1 - \rho\right).$$

*Moreover, the same lower bound holds for  $\varepsilon_{\text{ca}}(C, \delta)$  throughout the integer-admissible subinterval*

$$\delta \in \left[1 - \rho - \frac{2}{N_\rho}, 1 - \rho - \frac{1}{n}\right].$$

*If  $q \geq 2n$ , the lower bound improves to  $\geq 2^{-42}$ . In particular,*

$$\delta_C^*(2^{-128}) \leq 1 - \rho - 2^{-9} \quad (\rho \in \{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}\}), \quad \delta_C^*(2^{-128}) \leq 1 - \rho - 2^{-10} \quad (\rho = \frac{1}{16}).$$

*Proof.* We verify the hypotheses of Theorem 4.1 with  $N = N_\rho$ . Divisibility:  $a = n/N_\rho$  divides  $k$  because  $k/a = \rho N_\rho \in \mathbb{Z}$  (equal to 512, 256, 128, 128 respectively);  $(1 - \rho)N_\rho \geq 128 \geq 3$ . For (1), the entropy bound  $\binom{N}{\ell} \geq 2^{N\text{H}_2(\ell/N)}/(N+1)$  together with the conservative evaluations of  $\text{H}_2$  recorded in Table 1 gives  $\log_2 \binom{N_\rho}{\rho N_\rho+2} \geq 552$  in all four cases, while

$$|B|\left(\frac{q}{k} + 1\right) \leq q\left(\frac{q}{k} + 1\right) \leq q^2 + q < 2^{512} + 2^{256} < 2^{513} \leq 2^{552},$$

so (1) holds with at least 39 bits to spare (and over 500 bits at rate  $\frac{1}{2}$ ). Since  $D \subseteq \mathbb{F}^\times$ , we have  $n \leq q - 1$ ; for fixed  $n$ , the function  $1 - n/q$  is minimized at  $q = n + 1$ , hence  $1 - n/q \geq 1/(n + 1)$ . Also  $n = k/\rho \leq 16k \leq 2^{44}$ . Therefore

$$\frac{1}{2k} \left(1 - \frac{n}{q}\right) \geq \frac{1}{2k(n + 1)} > 2^{-86}.$$

If  $q \geq 2n$  then  $1 - n/q \geq 1/2$ , giving the sharper  $1/(4k) \geq 2^{-42}$ . The gap is  $2/N_\rho = 2^{-9}$  for  $\rho \in \{1/2, 1/4, 1/8\}$  and  $2/N_\rho = 2^{-10}$  for  $\rho = 1/16$ . Theorem 4.1 gives the MCA lower bound throughout the stated sub-capacity interval, and gives the CA lower bound on the integer-admissible subinterval ending at  $1 - \rho - 1/n$ .  $\square$

| $\rho$ | $N_\rho$ | $\ell_2 = \rho N_\rho + 2$ | $H_2(\ell_2/N_\rho) \geq$ | $\log_2 \binom{N_\rho}{\ell_2} \geq$ | needed | gap $2/N_\rho$ |
|--------|----------|----------------------------|---------------------------|--------------------------------------|--------|----------------|
| 1/2    | 1024     | 514                        | 0.99998                   | 1013                                 | 513    | $2^{-9}$       |
| 1/4    | 1024     | 258                        | 0.8143                    | 823                                  | 513    | $2^{-9}$       |
| 1/8    | 1024     | 130                        | 0.5490                    | 552                                  | 513    | $2^{-9}$       |
| 1/16   | 2048     | 130                        | 0.34105                   | 687                                  | 513    | $2^{-10}$      |

Table 1: Numerics for Corollary 5.1:  $\log_2 \binom{N}{\ell} \geq NH_2(\ell/N) - \log_2(N + 1)$ , rounded down. The “needed” column is the universal upper bound  $\log_2(|B|(q/k + 1)) < 513$  valid throughout the envelope  $q < 2^{256}$ ,  $k \geq 1$ .

*Remark 5.2* (hypothesis  $N_\rho \mid n$ ). For the 2-adic domains of the challenge this holds whenever  $n \geq 2^{10}$  (resp.  $2^{11}$ ), i.e. for all but very small instances. For mixed-radix smooth  $n$ , replace  $N_\rho$  by any divisor  $N \mid n$  of comparable size with  $\rho N \in \mathbb{Z}$ ; the entropy count in Table 1 is insensitive to the exact value of  $N$  at fixed  $\log_2 N$ .

*Remark 5.3* (two-tier answer to the grand challenge). Combining Corollary 5.1 with the grand-challenge cap of [Cho26b, Prop. “Grand-challenge cap”] (gap  $\frac{1}{64}$ , prime fields up to  $\approx 2^{150}$ , via value-set coverage) yields the current state of the negative side for smooth multiplicative domains:

$$\delta_C^*(2^{-128}) \leq \begin{cases} 1 - \rho - \frac{1}{64}, & \mathbb{F} \text{ prime, } q \lesssim 2^{150}, \\ 1 - \rho - 2^{-9}, & \rho \in \{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}\}, \text{ every } \mathbb{F}, q < 2^{256}, \\ 1 - \rho - 2^{-10}, & \rho = \frac{1}{16}, \text{ every } \mathbb{F}, q < 2^{256}. \end{cases}$$

Together with the positive results collected in [ABF26, Table 1], this pins the grand MCA threshold between the Johnson region and the explicit capacity-edge caps above, throughout the stated challenge envelope. The second and third tiers supersede the conditional cap of [Cho26b, Thm. “Universal cap”] ( $\delta^* < 1 - \rho - 2^{-11}$  under  $|\mathbb{F}| \geq 2n$ ), which was the previous state in this band.

**Corollary 5.4** (deployed parameters: KoalaBear sextic). *Let  $B = \mathbb{F}_p$  with  $p = 2^{31} - 2^{24} + 1$ , let  $\mathbb{F} = \mathbb{F}_{p^6}$  (so  $q = p^6 \approx 2^{185.93}$ ), let  $D \subseteq B^\times$  be the subgroup of order  $n = 2^{21}$ , and let  $k = 2^{20}$ ,  $\rho = \frac{1}{2}$ , as in [ABF26, §6.3]. Then*

$$\varepsilon_{\text{mca}}(\text{RS}[\mathbb{F}_{p^6}, D, 2^{20}], \delta) > (1 - 2^{-164}) 2^{-21} > 2^{-22}$$

for every

$$\delta \in \left[ \frac{1}{2} - 2^{-7}, \frac{1}{2} \right).$$

Moreover, the same lower bound holds for  $\varepsilon_{\text{ca}}$  throughout the integer-admissible subinterval

$$\delta \in \left[ \frac{1}{2} - 2^{-7}, \frac{1}{2} - 2^{-21} \right).$$

*Proof.* Apply Theorem 4.1 with  $N = 256$ ,  $a = 2^{13} \mid k$ ,  $(1 - \rho)N = 128 \geq 3$ , gap  $2/N = 2^{-7}$ . Hypothesis (1):  $\log_2 \binom{256}{130} \geq 256 \cdot H_2(130/256) - \log_2 257 \geq 256 \cdot 0.99982 - 8.01 > 247$ , while  $|B|(q/k + 1) < 2^{31}(2^{166} + 1) < 2^{198}$ . The error bound is  $\frac{1}{2^k}(1 - n/q)$  with  $n/q = 2^{21}/p^6 < 2^{-164}$ . Theorem 4.1 gives the stated MCA bound on  $[\frac{1}{2} - 2^{-7}, \frac{1}{2})$  and the stated CA bound on  $[\frac{1}{2} - 2^{-7}, \frac{1}{2} - 2^{-21})$ .  $\square$

**Corollary 5.5** (all interleaved rows of [ABF26, Tables 2–3]). *In the setting of Corollary 5.4, for each  $s = 2^j$  with  $0 \leq j \leq 12$  let  $C_s := \text{RS}[\mathbb{F}_{p^6}, D_s, k_s]$  with  $|D_s| = n_s = 2^{21}/s$  and  $k_s = 2^{20}/s$  (rate  $\frac{1}{2}$ ,  $s n_s = 2^{21}$  as deployed). Then for every  $\delta \in [\frac{1}{2} - 2^{-7}, \frac{1}{2})$ ,*

$$\varepsilon_{\text{mca}}(C_s^{\equiv s}, \delta) > \frac{s}{2^{21}}(1 - 2^{-164}) \geq 2^{-21}(1 - 2^{-164}).$$

Moreover, the same lower bound holds for  $\varepsilon_{\text{ca}}(C_s^{\equiv s}, \delta)$  on the integer-admissible subinterval

$$\delta \in \left[ \frac{1}{2} - 2^{-7}, \frac{1}{2} - \frac{1}{n_s} \right).$$

Every row of the deployed parameter family therefore fails the  $2^{-128}$  target at gap  $2^{-7}$  below capacity.

*Proof.* For each  $s \leq 2^{12}$ :  $256 \mid n_s$  (as  $n_s \geq 2^9$ ),  $a_s = n_s/256 = 2^{13}/s$  divides  $k_s = 2^{20}/s$ , and  $(1 - \rho) \cdot 256 = 128 \geq 3$ . Hypothesis (1) for  $C_s$ :  $|B|(q/k_s + 1) < 2^{31}(2^{166}s + 1) < 2^{198}$   $s \leq 2^{210} \leq 2^{247} \leq \binom{256}{130}$ . Theorem 4.1 gives the MCA lower bound for  $C_s$  throughout  $[\frac{1}{2} - 2^{-7}, \frac{1}{2})$ , and the CA lower bound throughout  $[\frac{1}{2} - 2^{-7}, \frac{1}{2} - 1/n_s]$ . Lemma 3.9 transfers the corresponding lower bounds to the  $s$ -interleaved code.  $\square$

**Proposition 5.6** (independent slacked variant via Theorem 2.8). *In the setting of Theorem 4.1, assume instead  $(1 - \rho)N \geq 2$ ,  $n \geq 3N$ , and*

$$\binom{N}{\rho N + 1} \geq |B| \cdot q.$$

Then

$$\varepsilon_{\text{ca}}\left(C, 1 - \rho - \frac{1}{N} + \frac{2}{n}, 1 - \rho - \frac{1}{n}\right) \geq \frac{1}{2n}.$$

In particular, with  $N = N_\rho$  as in Corollary 5.1 and  $n \geq 3N_\rho$ , every challenge-envelope instantiation satisfies this slacked CA lower bound, and  $\frac{1}{2n} \geq 2^{-45} \gg 2^{-128}$ .

*Proof.* Set  $\delta_0 := 1 - \rho - \frac{1}{N}$ . By  $(1 - \rho)N \geq 2$ ,  $\delta_0 \in (0, 1 - \rho)$ . By  $n \geq 3N$ , we also have  $\delta_0 + 2/n \leq 1 - \rho - 1/n$ , so the parameters in Theorem 2.8 are admissible. By Lemma 3.1(i), some  $u_z$  carries at least  $\binom{N}{\rho N + 1}/|B| \geq q$  codewords of  $C$  within radius  $\delta_0$ , so  $\Lambda(C, \delta_0) \geq q$ . The contrapositive of Theorem 2.8 gives

$$\varepsilon_{\text{ca}}\left(C, \delta_0 + \frac{2}{n}, 1 - \rho - \frac{1}{n}\right) \geq \frac{1}{2n}.$$



For the numerical claim, the worst case of  $\log_2 \binom{N_\rho}{\rho N_\rho + 1}$  over the four rates is at  $\rho = 1/8$ , where the entropy bound used in Table 1 certifies  $\log_2 \binom{1024}{129} \geq 1024 H_2(129/1024) - \log_2 1025 \geq 549$  (the exact value is  $\approx 554.7$ ); hence  $\binom{N_\rho}{\rho N_\rho + 1} \geq 2^{549} > 2^{512} > |B|q$  throughout  $q < 2^{256}$ . Finally  $n = k/\rho \leq 16k \leq 2^{44}$ , so  $1/(2n) \geq 2^{-45}$ .  $\square$

*Remark 5.7.* Proposition 5.6 is logically independent of Theorem 2.6 and yields a weaker error ( $\frac{1}{2n}$  versus  $\frac{1}{2k}$ ) in the *maximally slacked* form of correlated agreement — the internal radius sits at capacity minus  $\frac{1}{n}$  — which is the weakest failure mode one can certify. That even this fails by 83 orders of magnitude (in bits) above the  $2^{-128}$  target gives a separate fallback route through the imported BCHKS/ABF theorem.

## 6 Subfield confinement and the shape of certifying lines

The deployed instantiations run the protocol over an extension  $\mathbb{F}/B$  while keeping the domain in the base:  $D \subseteq B$ . The subfield-confinement theorem of [Cho26b, Thm. “Subfield confinement”] pins down where bad slopes can live for lines whose words are base-rational: all support-wise MCA-bad slopes lie in  $B$ . The following lemma refines this by tracking two radii, so that it also covers the proximity-loss form of correlated agreement (Definition 2.1); part (b) is the statement of [Cho26b], reproved here for completeness in the present notation.

**Lemma 6.1** (subfield confinement; refines [Cho26b]). *Let  $B \subseteq \mathbb{F}$  be finite fields,  $D \subseteq B^\times$ ,  $C = \text{RS}[\mathbb{F}, D, k]$ , and let  $f, g \in B^n \subseteq \mathbb{F}^n$  be  $B$ -valued words. Then:*

- (a) *for any  $0 \leq \delta_{\text{fld}} \leq \delta_{\text{int}} < 1$ , every CA-bad slope for  $(f, g)$  at radii  $(\delta_{\text{fld}}, \delta_{\text{int}})$  lies in  $B$ ;*
- (b) *for any  $\delta \in (0, 1)$ , every MCA-bad slope for  $(f, g)$  at radius  $\delta$  lies in  $B$ .*

*Consequently the bad-slope set of any  $B$ -valued pair has density at most  $|B|/q$  in  $\mathbb{F}$ . The same holds for pairs  $(\lambda f, \lambda g)$  with  $\lambda \in \mathbb{F}^\times$  and  $f, g$   $B$ -valued, by  $\mathbb{F}$ -linearity of  $C$ .*

*Proof.* Fix a  $B$ -basis  $1 = \beta_0, \beta_1, \dots, \beta_{m-1}$  of  $\mathbb{F}$ . Every word  $w \in \mathbb{F}^n$  decomposes uniquely as  $w = \sum_i \beta_i w_i$  with  $w_i \in B^n$ , and every codeword  $c \in \text{RS}[\mathbb{F}, D, k]$  decomposes as  $c = \sum_i \beta_i c_i$  with  $c_i \in \text{RS}[B, D, k]$ : write the coefficient vector of the underlying polynomial in the basis and evaluate at  $D \subseteq B$ .

Let  $z \in \mathbb{F} \setminus B$ , say  $z = \sum_i z_i \beta_i$  with  $z_{i^*} \neq 0$  for some  $i^* \geq 1$ . We first record the support-wise consequence. Suppose that  $c \in C$  agrees with  $f + zg$  on some set  $S \subseteq D$ . Since  $f_j, g_j \in B$ , the basis components of  $(f + zg)_j$  are  $f_j + z_0 g_j$  (component 0) and  $z_i g_j$  (components  $i \geq 1$ ). Agreement on  $S$  holds componentwise, so on  $S$ :

$$f + z_0 g = c_0, \quad z_{i^*} g = c_{i^*}.$$

Hence  $g$  agrees on  $S$  with  $z_{i^*}^{-1} c_{i^*} \in C$ , and then  $f = (f + z_0 g) - z_0 g$  agrees on  $S$  with  $c_0 - z_0 z_{i^*}^{-1} c_{i^*} \in C$ . Thus every support on which the line point  $f + zg$  is explained by  $C$  also explains the pair  $(f, g)$  on that same support.

For CA, if  $\Delta(f + zg, C) \leq \delta_{\text{fld}}$ , choose such an  $S$  with  $|S| \geq (1 - \delta_{\text{fld}})n$ ; the preceding paragraph gives  $\Delta_2((f, g), C^{\equiv 2}) \leq \delta_{\text{fld}} \leq \delta_{\text{int}}$ , so  $z$  is not CA-bad. For MCA, the preceding paragraph rules out every possible bad support  $S$  in the support-wise definition. Hence  $z$  is not MCA-bad either.  $\square$

**Corollary 6.2** (certifying lines over extensions are genuinely  $\mathbb{F}$ -valued). *At the parameters of Corollary 5.4,  $|B|/q = p^{-5} < 2^{-154}$ . Hence any pair  $(f, g)$  whose CA- or MCA-bad-slope density at some radius*

$$\delta \in \left[ \frac{1}{2} - 2^{-7}, \frac{1}{2} \right)$$

*exceeds  $2^{-154}$  contains a word that is not  $B$ -valued, even after removing a common scalar factor. In particular, any pair attaining the MCA lower bound  $> 2^{-22}$  of Corollary 5.4 is genuinely  $\mathbb{F}$ -valued. On the integer-admissible subinterval*

$$\delta \in \left[ \frac{1}{2} - 2^{-7}, \frac{1}{2} - 2^{-21} \right),$$

*the same statement applies to pairs attaining the CA lower bound. All explicit failure witnesses currently known on these domains (the locator lines of [Cho26a, Cho26b] and the fiber words of Lemma 3.1) are  $B$ -rational and are therefore inert here: the failure certified by Corollary 5.4 is fiber-borne and, at present, non-constructive.*

*Proof.* Immediate from Lemma 6.1 and Corollary 5.4: a  $B$ -valued pair, up to common scalar, has CA- and MCA-bad-slope density at most  $|B|/q = p^{-5} < 2^{-154} < 2^{-22}$ . Thus any pair witnessing the deployed lower bound cannot be of this form.  $\square$

*Remark 6.3.* Lemma 6.1 and Corollary 6.2 sharpen the division of labor between the two grand challenges of [ABF26] over deployed extensions: the *list* side does not deflate  $(\text{RS}[B, D, k] \subseteq \text{RS}[\mathbb{F}, D, k])$ , so all base-field list lower bounds, including Lemma 3.1, persist verbatim), while every sub- $2^{-128}$  MCA attack by explicit base-rational witnesses is killed by confinement — and yet MCA still fails at  $2^{-22}$ , through lines whose existence Theorem 2.6 extracts non-constructively from the surviving list mass. The deflation and decoupling corollaries of [Cho26b] record the same division at the no-loss radius, using the same corrected rounding  $p^{-5} = 2^{-154.94\dots} < 2^{-154}$  as in Corollary 6.2. Exhibiting such lines explicitly is Open Problem 8.1.

## 7 Prize-facing status and a completion program

The Proximity Prize asks for the largest admissible radius, not only for a near-capacity obstruction. Thus the present paper should be read as a one-sided theorem: it proves a universal unsafe band, while the remaining work is to prove a matching safe band below that cap, or to find additional obstruction floors that push the threshold even farther from capacity.

### 7.1 What is already settled by this paper

For the grand MCA challenge at target error  $\varepsilon^* = 2^{-128}$ , the unconditional output of Corollary 5.1 is the following negative certificate: for every challenge-envelope row satisfying the printed divisor hypothesis,

$$\delta \geq 1 - \rho - 2^{-9} \quad (\rho \in \{1/2, 1/4, 1/8\})$$

and

$$\delta \geq 1 - \rho - 2^{-10} \quad (\rho = 1/16)$$

are impossible MCA radii. Equivalently, any full determination of  $\delta_C^*(2^{-128})$  for these smooth multiplicative rows must search below these gaps. This conclusion uses only the self-contained

deep-point conversion, the quotient-fiber pigeonhole, and support-wise MCA monotonicity; it no longer rests on importing the Crites–Stewart conversion.

The theorem also distinguishes two different grades of information. First, the *threshold grade* is already enough for the prize inequality: the certified error is  $\gg 2^{-128}$ . Second, the *failure-profile grade* is still weak: the present proof gives error of order  $1/k$ , not error  $1 - o(1)$ . Closing the error-one profile at fixed gaps such as  $1/64$  is a different, stronger problem. It would improve our understanding of the failure landscape, but is not needed to obtain the stated upper bound on  $\delta_C^*(2^{-128})$ .

## 7.2 Integer-staircase formulation

Let

$$a(\delta) := \lceil (1 - \delta)n \rceil, \quad r(\delta) := n - a(\delta)$$

be the agreement and integer-radius forms of the same radius. For a fixed finite-slope line field  $F_{\text{line}}$ , write  $q_{\text{line}} = |F_{\text{line}}|$  and let

$$B_{\text{mca}}(a)$$

be the exact maximum number of MCA-bad line parameters at agreement at least  $a$ , for the line family used in the challenge. Then

$$\varepsilon_{\text{mca}}(C, 1 - a/n) = B_{\text{mca}}(a)/q_{\text{line}}$$

under the closed finite-slope convention. For projective-slope or curve samplers, the same staircase formulation applies after replacing  $q_{\text{line}}$  and  $B_{\text{mca}}$  by the corresponding sampler denominator and numerator. Thus the MCA prize problem is an integer staircase problem: find the first agreement  $a$  for which

$$B_{\text{mca}}(a) \leq 2^{-128} q_{\text{line}}.$$

The present paper supplies lower bounds on  $B_{\text{mca}}(a)$  near capacity by routing quotient-fiber list mass through deep points.

**Proposition 7.1** (one-step threshold certificate). *Fix a row  $C$  and a denominator  $q_{\text{line}}$ . Suppose that for some agreement  $a_0$  one has a lower certificate*

$$B_{\text{mca}}(a_0) > 2^{-128} q_{\text{line}}$$

*and an upper theorem*

$$B_{\text{mca}}(a_0 + 1) \leq 2^{-128} q_{\text{line}}.$$

*Then, under the closed integer-radius convention, radius  $1 - a_0/n$  is unsafe and radius  $1 - (a_0 + 1)/n$  is safe. As a real-radius supremum, the threshold is  $1 - a_0/n$  and is not attained if the closed ball at that endpoint is counted.*

*Proof.* The function  $B_{\text{mca}}(a)$  is nonincreasing in  $a$ , since raising the required agreement only removes witnesses. At the closed radius  $1 - a/n$ , the MCA error is  $B_{\text{mca}}(a)/q_{\text{line}}$ . The two displayed inequalities are therefore exactly the unsafe/safe transition at consecutive integer agreement levels.  $\square$

### 7.3 Conditional MCA completion theorem

The following theorem is a deliberately explicit proof blueprint. Each hypothesis is a concrete mathematical task, not a black-box “large field” assumption.

**Theorem 7.2** (conditional MCA threshold theorem). *Fix a smooth multiplicative row  $C = \text{RS}[F, D, k]$ , a line field  $F_{\text{line}}$ , and a target  $\varepsilon^* = 2^{-128}$ . Suppose the following four inputs are proved for this row, uniformly in the received line:*

- (i) **Quotient-profile lower and upper ledger.** *For every divisor  $c \mid n$  and every agreement  $a = mc + s$  with  $0 \leq s < c$ , the quotient-periodic bad slopes, including remainder supports consisting of  $m$  full fibers plus  $s$  residual points, are counted with overlaps across divisors accounted for. Write the resulting divisor-lattice upper numerator as  $B_Q^{\text{all}}(a)$ ; individual divisor contributions  $B_Q(a; c)$  may also be printed, but the upper ledger must count their union or a certified safe sum rather than only their maximum.*
- (ii) **Aperiodic residue-line packing.** *After removing tangent and quotient-periodic locators, the Hankel/residue-line incidence variety has at most  $B_{\text{ap}}(a)$  bad line parameters.*
- (iii) **Extension-line accounting.** *If  $F_{\text{line}}$  is a proper extension of the field generated by  $D$ , either all genuinely extension-valued line witnesses are included in  $B_{\text{ap}}(a)$ , or a separate extension term  $B_{\text{ext}}(a)$  is proved.*
- (iv) **No missing line family.** *The line family in the theorem is the same finite-slope or projective-slope family used by the challenge or protocol under consideration.*

Assume moreover that the resulting upper numerator

$$B^+(a) := B_{\text{tan}}(a) + B_Q^{\text{all}}(a) + B_{\text{ap}}(a) + B_{\text{ext}}(a)$$

and a corresponding lower numerator  $B^-(a)$  satisfy

$$B^-(a_0) > \varepsilon^* q_{\text{line}}, \quad B^+(a_0 + 1) \leq \varepsilon^* q_{\text{line}}$$

for some  $a_0$ . Then  $\delta_C^*(\varepsilon^*)$  is determined at the one-step accuracy of Proposition 7.1.

*Proof.* The four hypotheses classify every possible MCA-bad line parameter into one of four buckets: tangent, quotient-periodic, aperiodic, or genuinely extension-line. The quotient bucket is counted by the divisor-lattice union ledger  $B_Q^{\text{all}}$ , so overlaps or simultaneous quotient descriptions cannot invalidate the upper bound. The displayed  $B^+(a)$  is therefore an upper bound for  $B_{\text{mca}}(a)$ , while  $B^-(a)$  is a certified lower bound. The conclusion is exactly Proposition 7.1.  $\square$

The present paper supplies part of the lower numerator  $B^-(a)$  in the near-capacity band: quotient-fiber list mass forces MCA mass above the printed universal caps. It does not prove the aperiodic upper input (ii), and therefore does not by itself determine the exact staircase.

### 7.4 Conditional interleaved-list completion theorem

The grand list-decoding challenge asks for the largest  $\delta$  such that an interleaved list, divided by the field used for the relevant challenge or list ledger, is below  $2^{-128}$ . In the simplest same-field form this is

$$|\Lambda(C^{\equiv m}, \delta)| \leq 2^{-128} |F|.$$

The quotient-fiber construction in this paper already gives lower floors for ordinary lists, and ordinary lists embed into interleaved lists by placing the same received word in one row and zero words in the others. The missing positive direction is an arbitrary-word locator-fiber theorem.

**Theorem 7.3** (conditional interleaved-list threshold theorem). *Fix an interleaving arity  $m$ . Suppose that for every received word  $U$  and agreement  $a = k + \sigma$  one has a base-code list bound of the form*

$$|\Lambda(C, 1 - a/n, U)| \leq L_Q(a, U) + L_{\text{ap}}(a),$$

*where  $L_Q$  is an explicit quotient-profile list contribution and  $L_{\text{ap}}$  is a uniform aperiodic locator-fiber bound. Suppose also that the same bound is available at the higher agreement levels that occur in the codegree recursion, starting with  $2a - k$ . Then the interleaved-list threshold is obtained by applying the codegree recursion row by row and comparing the resulting numerator to  $2^{-128}|F|$ .*

*Proof.* For two rows, split the second-row list according to whether its agreement set has size below or above  $2a - k$ . Below  $2a - k$ , two first-row codewords would agree on more than  $k$  positions and hence are equal; above  $2a - k$ , use the base-code list bound at agreement  $2a - k$ . Iterating gives the stated  $m$ -row recursion. This is the same codegree mechanism used in the experimental ledger; the only missing input is the arbitrary-word locator-fiber bound.  $\square$

## 7.5 Concrete proof obligations

The fastest route from the present cap to a full prize solution is to turn the following proof obligations into theorems. The lower-floor half of T1 is now partly supplied by the quotient-remainder prefix lemma of Lemma 3.5; its upper/divisor-union half remains open.

- T1. Remainder quotient-profile theorem.** Generalize the fiber lemma from full quotient-fiber unions to all agreements  $a = mc + s$ . The target is an exact lower and upper quotient ledger over the divisor lattice of  $D$ , with intersections between divisor scales handled by an exact union count or by a certified safe sum, and with the subfield denominator  $|B|$  printed separately from the line denominator  $q$ . This is the most direct extension of the present paper.
- T2. Aperiodic Hankel-packing theorem.** Use the syndrome/Hankel normal form of [Cho26b] to prove that, after quotient-periodic and tangent locators are removed, the number of remaining bad line parameters is at most  $B_{\text{ap}}(a)$ , preferably polynomial in  $n$  above the corrected reserve. This is the main MCA upper-bound problem.
- T3. Extension-line lift or counterexample.** For  $B \subset F$  and  $D \subset B$ , either prove that genuinely  $F$ -valued lines are covered by the aperiodic ledger with a stable numerator, or exhibit explicit extension-valued counterexamples. Corollary 6.2 shows that deployed failures certified by this paper must be genuinely extension-valued; this is not optional protocol bookkeeping.
- T4. Arbitrary-word locator-fiber theorem.** Prove the base-code list bound needed by Theorem 7.3, first for monomial-prefix received words and quotient-free dimensions, then for arbitrary received words with quotient cores budgeted.

**T5. Curve and projective-line ledgers.** Extend the affine-line Hankel classification to projective slopes and to finite-parameter power curves  $f_0 + \gamma f_1 + \dots + \gamma^d f_d$ . The fixed-locator equation becomes a vector polynomial of degree at most  $d$  in  $\gamma$ ; the expected loss is a factor of  $d$  plus exceptional vanishing-vector strata.

**T6. Certificate scanner.** Implement a finite staircase scanner that, for each challenge row, prints tangent, quotient, generated-field entropy, aperiodic, extension, curve, and interleaved-list numerators at every agreement  $a$ . The desired output is an interval

$$a_{\text{unsafe}} < a_C^* \leq a_{\text{safe}}$$

that shrinks to one step when Theorem 7.2 or Theorem 7.3 applies.

**T7. Formal finite core.** Formalize the self-contained deep-point conversion, the quotient-fiber pigeonhole, support-wise MCA monotonicity, and the integer staircase lemma. These are small enough for proof-assistant checking and directly strengthen a prize submission.

## 7.6 Recommended submission posture

As a standalone prize-facing submission, this paper should claim a significant partial result: a universal negative band for the grand MCA challenge over smooth multiplicative rows in the stated field-size envelope. It should not claim the full prize threshold, a positive theorem below the cap, protocol soundness, or a curve-MCA theorem. The full prize claim should be made only after the conditional ledgers above are replaced by theorems and the staircase scanner produces a one-step unsafe/safe transition for the relevant rows.

## 8 Discussion

**Values versus fibers.** Above the per-rate value-set reach of [Cho26b] (roughly  $2^{150}$ – $2^{162}$ ), the value route — showing that the slope multiset  $e_1(\ell^\wedge Q)$  covers a  $2^{-128}$  fraction of  $\mathbb{F}$  at a fixed large prime — runs into the per-fiber collision problem ([Cho26b, Rem. “The remaining band”]): characteristic-zero locator values can collide arbitrarily badly under reduction at a worst-case prime. [Cho26b] therefore closed the band at the  $\delta^*$  grade only conditionally and with weaker constants — error  $2^{-42}$  at gap  $2^{-11}$ , assuming  $|\mathbb{F}| \geq 2n$ , via quotient-core lists and the same conversion Theorem 2.6 — leaving the *error-one* grade open. The present route also never counts values, and improves each parameter of that cap: pigeonhole guarantees one heavy fiber regardless of how the values  $e_1(A)$  distribute, Theorem 2.6 converts heavy fibers into correlated-agreement error directly, and the subfield pigeonhole carries the construction to extension fields with no condition relating  $n$  and  $|\mathbb{F}|$ . The cost relative to the error-one results of [Cho26a, Cho26b] below  $2^{150}$  is quantitative: error  $\approx \frac{1}{2k}$  instead of  $1 - o(1)$ , and gap  $2^{-9}$  (uniformly  $2^{-10}$  across all four rates) instead of  $\frac{1}{64}$ . For the prize question — is  $\varepsilon_{\text{mca}} \leq 2^{-128}$  achievable above the stated gaps? — this cost is immaterial.

**Limits of the method.** Hypothesis (1) needs  $N H_2(\rho) \gtrsim 2 \log_2 q - \log_2 k$ , so the smallest certifiable gap is

$$\frac{2}{N} \approx \frac{2 H_2(\rho)}{2 \log_2 q - \log_2 k} \approx \frac{H_2(\rho)}{\log_2 q},$$

and the certified error is capped at  $\approx \frac{1}{k}$  by the shape of Theorem 2.6 (any  $\eta < 1$ ). Since  $q$  enters only as the size of the field from which the line challenge  $\gamma$  is drawn, this is also a protocol-facing floor: enlarging the challenge field cannot push a deep-point-conversion failure below gap  $\approx H_2(\rho)/\log_2 q_{\text{line}}$ , which quantitatively bounds the “larger challenge field” fallback in certificate frameworks. The construction also needs  $N \leq n$ , i.e.  $\log_2 q \lesssim H_2(\rho) n/2$ ; this covers the entire cryptographic regime  $\text{poly}(n) \leq q \leq 2^{o(n)}$ , but for  $q \geq 2^{H_2(\rho)n/2}$  the method is vacuous and the value-route caps of [Cho26a, Cho26b] are the only ones known. Within the band, the *error-one* question at gap  $\frac{1}{64}$  (Open Problem 8.2) remains exactly as open as before: the present result bounds  $\delta^*$ , not the failure profile near capacity.

**Comparison with the cyclotomic sieve.** The sieve of [Cho26a] produces, for infinitely many primes, failures of error  $(\log p)^{-6}$  at gap  $\Theta(1/\log p)$ , but its prime set is given by Siegel–Walfisz and is ineffective. Corollary 5.1 trades error strength for universality: every field below  $2^{256}$ , fully effective, and elementary given Theorem 2.6. The two results triangulate the failure landscape from different directions.

**Pressure on the packing conjecture.** By the exact normal form of [Cho26b],  $\varepsilon_{\text{mca}}(C, \delta) = \max_t \Lambda_t^{\text{NC}}(\delta)/q$ , where  $\Lambda^{\text{NC}}$  is a noncontained residue-line packing number. Corollary 5.4 therefore forces  $\max_t \Lambda^{\text{NC}} \geq (q-n)/2k \approx 2^{164}$  at gap  $2^{-7}$  on the deployed sextic — superpolynomially many noncontained residue lines at  $n = 2^{21}$ . This puts no direct pressure on the final MCA conjecture of [Cho26b, Conj. “Final floor- and quotient-corrected MCA asymptotic”]: at gap  $2^{-7}$  both of its corrected-reserve hypotheses fail ( $\sigma = 2^{14}$  lies below  $Cn/\log_2 n \approx 2^{16.6}$ , and  $\sigma \log_2 q_{\text{gen}} = 2^{14} \cdot 31 \approx 2^{19}$  is far below  $\log_2 \binom{n}{k+\sigma} \approx 2^{21}$ ), so the conjectured ceiling of the form  $(n^{1+o(1)} + 2^{(\beta/H_2) Q_{H_2}(\eta)})/q$  is simply not asserted there — the packing mass extracted through Theorem 2.6 lives in the region the framework already marks as below-floor. The genuine structural question this note raises for the framework of [Cho26b] is whether conversion-extracted fiber mass persists *above* the corrected reserve: whether, at reserves where the conjecture does apply, the fiber route still produces packings beyond the  $n^{1+o(1)}$  aperiodic term plus the quotient-periodic term, which would force an additional fiber-borne term into the conjectured ceiling. This is open.

**Verification caveat.** The main cap no longer depends on importing the Crites–Stewart theorem: Theorem 2.6 proves the needed deep-point list-to-CA conversion directly, including the integer-radius condition  $f \leq n-k-1$ . Theorem 2.8 remains a separate slacked fallback imported through the BCHKS/ABF route, and readers using that fallback should consult the underlying source directly. The main composition is deliberately shallow: it uses the self-contained conversion, the locator-fiber pigeonhole, and support-wise MCA monotonicity.

## Open problems.

**Open Problem 8.1.** Exhibit explicit pairs  $(f, g)$  over  $\mathbb{F}_{p^6}$  (KoalaBear sextic, parameters of Corollary 5.4) with MCA-bad-slope density  $> 2^{-22}$  at gap  $2^{-7}$ . On the integer-admissible subinterval, ask for the corresponding CA-bad-slope density as well. By Corollary 6.2 such pairs exist and are not  $B$ -rational; the natural candidates are residue-line normal forms of [Cho26b] with denominator  $E \in \mathbb{F}[X]$  not defined over  $B$ .



**Open Problem 8.2.** Error one in the band: does  $\varepsilon_{\text{mca}}(C, 1 - \rho - \frac{1}{64}) = 1 - o(1)$  hold for all primes  $2^{150} \leq p \leq 2^{256}$ ? This is the per-fiber collision problem proper; Corollary 5.1 caps  $\delta^*$  there but says nothing above error  $\approx \frac{1}{2k}$ .

**Open Problem 8.3.** Improve the threshold of Theorem 2.6: any strengthening of the hypothesis range  $\varepsilon_{\text{ca}} \leq \eta/k$  toward  $\varepsilon_{\text{ca}} \leq \eta \cdot \text{polylog}(q)/k$ , or of the list conclusion, immediately tightens the certified error of Theorem 4.1 toward  $1/\text{poly log}$ ; the gap, by contrast, is governed by the binomial in (1) and would need a denser fiber construction to shrink.

**Open Problem 8.4.** Beyond multiplicative smoothness: Lemma 3.1 uses only that  $D$  is a union of complete fibers of a low-degree map with lacunary power structure. Carry the composition to isogeny-smooth domains — in particular the circle-group domains of Circle STARKs over  $p = 2^{31} - 1$ , where Chebyshev locators  $\prod_b (T_{2^j}(x) - x_b)$  play the role of  $\prod_b (X^a - b)$  — to obtain the same universal cap for that deployed family.

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